

Complete Solutions Worked Example

Concise revision notes for reading free variables, special solutions, and $x = x_p + x_n$.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. A system with free variables has infinitely many solutions. One particular solution solves $Ax = b$; special solutions describe all movement inside $N(A)$.

The final answer has the form

$$x = x_p + x_n.$$

If $s_1, \dots, s_k$ are special solutions, then

$$x = x_p + c_1 s_1 + \dots + c_k s_k.$$

2 Reading Free Variables

Free variables come from columns without pivots. In a reduced system, set each free variable one at a time to create nullspace directions.

Variable type	Meaning
Pivot variable	Forced by the equations.
Free variable	Chosen freely; creates a nullspace direction.
Particular choice	Usually set free variables to simple values to solve $Ax = b$.

3 Particular Solution

For a consistent system

$$Ax = b,$$

choose convenient free variable values, often zeros, and solve for pivot variables. This gives one particular solution

$$x_p.$$

It is not the whole answer unless the nullspace is zero.

4 Special Solutions

To find the nullspace, solve

$$Ax = 0.$$

For each free variable:

1. set that free variable to 1;
2. set the other free variables to 0;
3. solve for pivot variables.

The resulting vectors $s_1, \dots, s_k$ form a basis for $N(A)$:

$$N(A) = \text{span}\{s_1, \dots, s_k\}.$$

5 Putting the Answer Together

If one particular solution is x_p and the nullspace basis is s_1, s_2 , write

$$x = x_p + c_1 s_1 + c_2 s_2.$$

The constants c_1, c_2 are arbitrary.

Meaning. The particular solution reaches b . The nullspace part moves without changing Ax .

6 Common Mistakes

Mistake	Correct exam habit
Giving only x_p .	Add the full nullspace.
Using $Ax = b$ to find special solutions.	Use $Ax = 0$.
Multiplying x_p by arbitrary constants.	Constants multiply only nullspace directions.
Forgetting one free variable.	Number of special solutions equals number of free variables.
Choosing complicated free values.	Use 0's and 1's unless the system suggests otherwise.

7 Final Exam Checklist

Checklist.

1. Row-reduce the augmented matrix $[A \ b]$.
2. Check consistency.
3. Mark pivot variables and free variables.
4. Choose simple free-variable values to get x_p .
5. Solve $Ax = 0$ for special solutions.
6. Create one special solution per free variable.
7. Write $x = x_p + c_1s_1 + \cdots + c_ks_k$.
8. Check by substituting: $Ax_p = b$ and $As_i = 0$.